
FOINWI

Grow Beyond Numbers

GST CALCULATOR

CURRENT-RULE & PRODUCT ACCURACY RECORD

Internal Governance Record

Audit / Review Period	10 September 2026 (read-only current-rule audit, then same-day correction)
Final Production Checkpoint	10 September 2026, 23:24:54 IST
Git Commit	3a43a2e 3a43a2e13ec9a5bc1071ebe19134124ccb1460db
Commit Message	Correct GST calculator for current rate framework
Branch	main

This document is an internal FOINWI governance and product-accuracy record. It is not marketing material, legal advice, a filing tool, or a certification. It does not claim that the calculator is legally certified, government approved, CA certified, regulator approved, or permanently accurate.

1. Purpose of Record

This internal record documents the completed FOINWI current-rule and product-accuracy review for this calculator only. It records calculator scope, the regulatory/current-rule review, formula review, assumptions, discrepancies discovered, corrections implemented, validation evidence, remaining limitations, authoritative sources used, and the Git production checkpoint.

- Record that GST arithmetic identities were already correct, and that the material P0 issue was current-rule framing (common rates capped at 28% after notified 40%).
- Record visitor-entered rate arithmetic, Rule 35 extract identity, and combined GST only.
- Preserve NOT DEFINITELY VERIFIED items from the GST audit.

2. Product Scope

What this calculator does

- Educational transaction-level GST arithmetic for a visitor-entered applicable GST rate.
- Add GST to taxable value: $GST = taxable \times rate / 100$; $invoice = taxable + GST$.
- Extract GST from a GST-inclusive amount using CGST Rules, 2017, Rule 35: $taxable = inclusive \times 100 / (100 + rate)$.
- Rate range 0-40% so 40% can be typed as illustration. Presets 0 / 5 / 18 / 28 / 40 are convenience choices, not a complete schedule.
- Combined GST component only. Internal values unrounded; display two decimal places as a product rule.

What this calculator does not do

- Not a rate/HSN/SAC classifier or place-of-supply adjudicator.
- Does not determine CGST/SGST/UTGST/IGST.
- Not a GSTR, ITC, composition, registration, cess, or e-invoice/e-way calculator.
- Does not decide exempt / nil-rated / zero-rated / outside GST from a 0% arithmetic result.
- FOINWI does not determine the legally applicable rate.

3. Pre-Correction State

- Math in GstCalculator.jsx was already the exclusive add identity and the Rule 35 inclusive identity. The inclusive bug ($inclusive \times rate / 100$) was not present.
- Copy described "common Indian GST rates" with a hard max 28% after notified combined rates include 40%, plus special 0.25% / 1.5% / 3%.
- Silent 18% default without "starting value only"; goods-vs-services beginner copy implied classification.
- No gstEngine / gstRules; calculateGst was not exported; validator used tautological hardcoded 10000/1800/11800.
- Slider could not enter 40% or 0.25%. Display rupee-rounding could desync paise.

4. Current-Rule Audit Findings

Audit bottom line: math is sound; current-rule trust was not. Uncertainties in audit section 25 are preserved.

P0 Stop describing the tool as covering common Indian GST rates while capping at 28% after 22 Sep 2025 notified rates include 40% and special goods rates.

P0 Do not imply FOINWI knows the visitor's applicable rate (default 18% + goods-vs-services classification copy).

P1 Central engine; visitor-entered rate to 40%; 18% starting value only; ITC/not-a-classifier/not-cess disclaimers; validator imports production; two-decimal display rule.

P2 Convenience chips 0/5/18/28/40 clearly not a complete schedule. 12% is not a chip.

Uncertainties preserved from the review

- Gazette G.S.R. number and full commencement clause of 09/2025-Central Tax (Rate) from a CBIC HTML page: NOT DEFINITELY VERIFIED. Newsletter + ITR twin + 15/2025 commencement were used instead.
- Complete residual 12% service list after 15/2025: not proven that none remain.
- Whether the tobacco/pan masala 40% + cess-off transition was later notified before September 2026: Council said later date; Schedule VII 28% still appeared in 9/2025-ITR.
- Current notified composition turnover as of Sep 2026: not verified.
- Any 2026 Central/Integrated Tax (Rate) notification rewriting Sep 2025 slabs: none found; still not a proof that none exists on an unindexed Gazette page.

5. Authoritative Basis

The sources below were used during the completed review. This section is a record of that review. It is not a fresh regulatory reinterpretation.

CBIC - CGST Act, 2017, s.9(1) - intra-State levy at notified rates not exceeding 20%

Dates: In force; retrieved during the audit

URL: https://taxinformation.cbic.gov.in/content/html/tax_repository/gst/acts/2017_CGST_act/active/chapter3/section9_v1.00.html

Supported: Levy framework; combined intra-State 40% is 20% CGST + 20% SGST/UTGST.

CBIC - IGST Act, 2017, s.5(1) - inter-State levy at notified rates not exceeding 40%

Dates: In force; retrieved during the audit

URL: https://taxinformation.cbic.gov.in/content/html/tax_repository/gst/acts/2017_IGST_Act/active/chapteriii/section5_v1.00.html

Supported: Supports why a 40% combined illustration is legally possible.

CBIC - CGST Rules, 2017, Rule 35 - tax amount from value inclusive of taxes

Dates: In force

URL: https://taxinformation.cbic.gov.in/content/html/tax_repository/gst/rules/cgst_rules/active/chapter4/rule35_v1.00.html

Supported: Extract-GST identity used by the production engine.

GST Council / PIB - 56th GST Council recommendation of merit 5%, standard 18%, de-merit 40%; asked implementation from 22 September 2025

Dates: 3 September 2025 (recommendation, not itself law)

URL: https://gstcouncil.gov.in/sites/default/files/2025-09/press_release_press_information_bureau_0.pdf

Supported: Council recommendation distinguished from notified law.

GST Council newsletter / CBIC notifications - Rate notifications dated 17 September 2025, generally effective 22 September 2025 (services confirmed in 15/2025-CTR commencement)

Dates: 17 September 2025; commencement 22 September 2025

URL: https://gstcouncil.gov.in/sites/default/files/2025-10/september_newsletter.pdf

Supported: Notified law used for the current-rate-framework framing of the calculator.

No later general rate overhaul was found as notified law through 10 September 2026. The 57th GST Council meeting had not sat.

6. Final Production Formula

Add mode:

$$\text{gstAmount} = \text{taxableValue} \times \text{gstRatePercent} / 100$$

$$\text{invoiceValue} = \text{taxableValue} + \text{gstAmount}$$

Extract mode (Rule 35 identity):

$$\text{taxableValue} = \text{inclusiveAmount} \times 100 / (100 + \text{gstRatePercent})$$

$$\text{gstAmount} = \text{inclusiveAmount} - \text{taxableValue}$$

Equivalent extract: $\text{inclusive} \times \text{rate} / (100 + \text{rate})$

NOT $\text{inclusive} \times \text{rate} / 100$

NaN / Infinity / negative amount or rate -> valid false; money fields 0

Variables

Variable	Meaning
taxableValue	Amount before GST in add mode
inclusiveAmount	GST-inclusive amount in extract mode
gstRatePercent	Visitor-entered applicable rate, 0-40 in this product

Timing and method assumptions

- Transaction-level arithmetic only; no tax period, no place of supply, no return period.
- Internal values stay unrounded. Display uses two decimal places; this is not statutory invoice rounding.

7. Final Rules / Assumptions

Material production assumptions are classified below. Statutory / official rule is distinguished from a FOINWI product assumption and from an illustrative user-editable assumption.

Class	Assumption
STATUTORY / OFFICIAL RULE	Add identity is exclusive-rate arithmetic. Extract identity is Rule 35 for a single combined rate.
FOINWI PRODUCT ASSUMPTION	Combined GST only; no CGST/SGST/IGST split; no cess; no classifier.
FOINWI PRODUCT ASSUMPTION	Rate chips 0/5/18/28/40 are illustration, not a complete schedule. 12% is enterable but not a chip.
FOINWI PRODUCT ASSUMPTION	Amount cap Rs 1 crore is a product limit, not a statutory GST threshold. Two-decimal display is a product rule.
ILLUSTRATIVE USER-EDITABLE ASSUMPTION	Applicable GST rate is supplied by the visitor. 18% is only the starting value.

8. Corrections Implemented

Correction 1

Old: "Common Indian GST rates" + max 28%; 18% default without starting-value caveat; goods vs services classification copy.

Corrected: Visitor-entered applicable rate 0-40%; 18% starting value only; FOINWI does not determine the legally applicable rate.

Reason: Current-rule framing after 22 September 2025 notified 40% de-merit.

Correction 2

Old: Inline calculateGst; validator hardcoded 10000 -> 1800 / 11800.

Corrected: gstRules + gstEngine; validate:gst imports calculateGstEstimate; 133 checks including inclusive-bug regression (Rs 1,180 @ 18% extract = GST 180, not 212.40).

Reason: Architecture and regression lock of the already-correct identity.

9. Deterministic Test Vectors

The vectors below were executed against the production engine in the calculator validator. No unexecuted tests are listed.

Inputs	Expected	Actual / validated	Result
A: add Rs 1,000 @ 18%	GST 180, invoice 1,180	180 / 1,180	PASS
B: add Rs 1,000 @ 5%	50 / 1,050	50 / 1,050	PASS
C: add Rs 1,000 @ 28%	280 / 1,280	280 / 1,280	PASS
D: add Rs 1,000 @ 40%	400 / 1,400	400 / 1,400	PASS
E: extract Rs 1,180 @ 18%	taxable 1,000, GST 180 (not 212.40)	GST 180	PASS
G: 0%	GST 0, invoice 1,000	0 / 1,000	PASS
I: Rs 999.99 @ 18% add	internal 179.9982 / 1179.9882; display Rs 180.00 / Rs 1,179.99	As correction report	PASS
0.25% / 1.5% / 3% on Rs 10,000 add	25 / 150 / 300	As correction report	PASS

10. Validation Evidence

Check	Recorded result
validate:gst	Passed 133 checks
validate:calculators	Passed
validate:trust	Passed 32 checks
validate:hardening	Passed 15 checks
lint	Passed
build	Passed
validate:guide / validate:learn	Passed (extra, recorded in the GST correction report)

Test limitations

- Browser click-through was not possible: the IDE browser and shell could not reach a live Vite server.
- This archive does not claim a later browser re-test.

11. Remaining Limitations

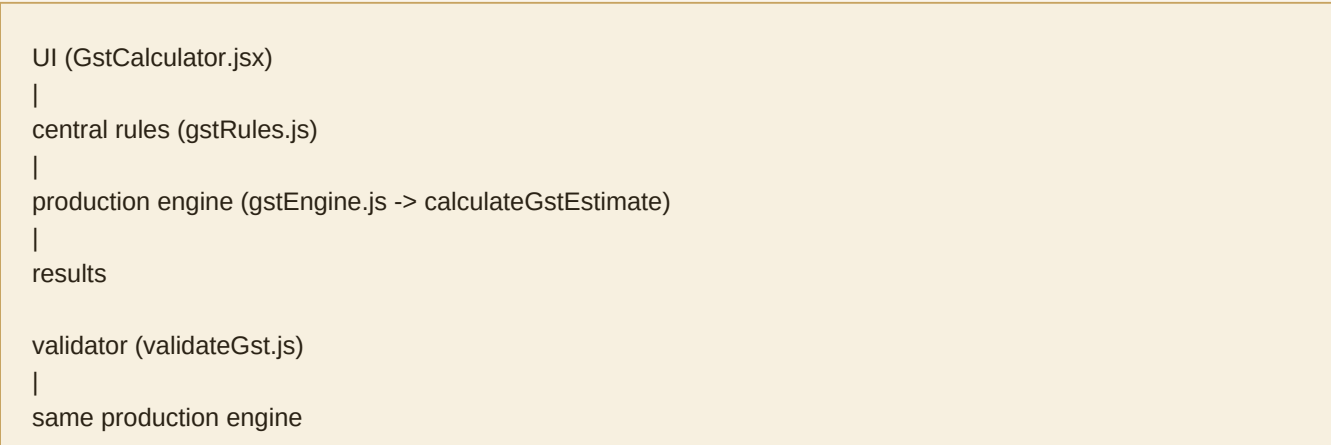
This section is mandatory. It records what this calculator deliberately does not model. It protects FOINWI from overclaiming.

- Combined GST only; no CGST/SGST/IGST split.
- No cess, composition, registration, ITC, HSN/SAC, or e-invoice logic.
- Rate chips are a small illustration set, not the statutory schedule.
- Rs 1 crore amount cap is a product limit.
- Two-decimal display is a product rule, not invoice law.
- 0% arithmetic does not classify exempt / nil-rated / zero-rated / outside GST.

12. Trust & User-Communication Standard

- Educational transaction-level arithmetic; visitor supplies the applicable rate.
- 18% is only the starting value, not FOINWI classification.
- Not return liability, not ITC, not cess.
- Presets are illustration, not a complete GST rate schedule.

13. Architecture



- Validation imports calculateGstEstimate rather than hardcoding 10000/1800/11800.

14. Change Record

Calculator	GST Calculator
Audit date	10 September 2026 (read-only current-rule audit, then same-day correction)
Correction date	10 September 2026, 23:24:54 IST
Final Git commit hash	3a43a2e13ec9a5bc1071ebe19134124ccb1460db
Commit message	Correct GST calculator for current rate framework
Branch	main
Relevant production files	src/utils/gst/gstRules.js; src/utils/gst/gstEngine.js; src/components/GstCalculator.jsx; scripts/validateGst.js; scripts/validateAllCalculators.js; package.json (validate:gst); src/utils/calculatorFormat.js (currencyPaise); src/components/ui/InputField.jsx (decimal keypad); src/components/ui/calculator-ui.css; Direct GST copy in calculators, explains, insights, journeys, Learn, Guide
Validator	scripts/validateGst.js

Validation command	npm run validate:gst
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Final Git commit 3a43a2e verified from git log. Pre-work HEAD was 5352acc (NPS correction). This is also HEAD of main at archive time.

15. Final Status

FOINWI INTERNAL PRODUCT ACCURACY STATUS

Checkpoint	Status
Current-rule audit completed	YES
Correction completed	YES
Production engine validated	YES
Regression checks passed	YES
Known limitations documented	YES

Status reflects the rules, assumptions, sources and production checkpoint documented in this record. Regulatory and statutory requirements may change after the recorded review date.

This record does not state that the calculator is 100% legally accurate, permanently compliant, government certified, or error-free forever.